

PROJECT PLAN

AI-Agent SOC Triage Engine

A 12-week build plan for an autonomous investigation agent that triages security alerts end to end.

1

End-to-end autonomous pipeline

Identity, SIEM, and EDR mock services feed a LangGraph investigation engine that reasons over each alert.

2

Built in 12 focused weekly sprints

From FastAPI foundations through correlation, MITRE ATT&CK mapping, AI reporting, and a live SOC dashboard.

3

A human stays in the loop

Every automated finding routes through an analyst approval gate before action is taken.



Built for a real SOC bottleneck

1

Analysts are drowning in alerts

Security teams triage thousands of alerts a day; most turn out benign. An agent that pre-investigates each one lets analysts spend their time on the threats that matter.

2

Builds in-demand agentic AI skills

The plan is a hands-on path through FastAPI, LangGraph, tool-using agents, correlation logic, and MITRE ATT&CK mapping — the core stack behind modern security automation.

3

Delivers a portfolio-grade system

The result isn't a toy demo: it's a working investigation engine with a dashboard, documentation, and a case study — proof of applied, end-to-end engineering.

HOW IT COMES TOGETHER

Three phases, twelve weeks

WEEKS 1–4

Foundation & Data Sources

- Alert schema + Identity API in FastAPI
- Synthetic identity data & PostgreSQL
- Mock SIEM and EDR services with attack events

WEEKS 5–7

Investigation Engine

- LangGraph supervisor & investigation state
- SIEM/EDR tools wired into the agent
- Correlation engine + MITRE ATT&CK mapping

WEEKS 8–12

Human Layer & Delivery

- AI-generated investigation reports
- Analyst approval workflow + Streamlit dashboard
- Full evaluation, docs, and portfolio case study

THE TAKEAWAY

One roadmap, one working agent

Twelve weeks turns raw alert data into an autonomous, explainable investigation — with a human approving every final call.

1

A complete triage pipeline

Identity, SIEM, and EDR data feed a LangGraph agent that investigates, correlates, and maps every alert to MITRE ATT&CK.

2

Human-in-the-loop by design

The approval gate and analyst dashboard keep a person in control of every action the agent recommends.

3

A finished, demonstrable system

Ends in Week 12 with a tested engine, technical documentation, a recorded demo, and a portfolio case study.